

Infosys Springboard Virtual Internship 7.0 Completion Report

Batch Number: 01

Start date: 29-06-2026

Internship Duration: 6 Weeks

Project Title: Talent Management Platform for Employee Performance and Career Growth

Internship Duration: 6 weeks

1. Project Title

Talent Management Platform for Employee Performance and Career Growth

2. Project Objective

The primary objective of this project is to engineer a privacy-aware, enterprise-grade AI platform designed to modernize talent management. The system architecture integrates multi-role access control, real-time performance analytics, and a sophisticated RAG (Retrieval-Augmented Generation) document search engine. Key functional pillars include automated AI exam generation, 6-day intensive study sprint orchestration, and the "Sphere" Voice AI Assistant. The platform is underpinned by a high-concurrency, lock-free SQLite WAL (Write-Ahead Logging) architecture to ensure data integrity and seamless performance under enterprise-level load.

3. Project Description in Detail

The platform serves as a unified ecosystem for employee development. At its core, the **Document Ingestion & RAG Engine** utilizes BGE-Large embeddings stored in ChromaDB to facilitate semantic search across sensitive enterprise documentation. This allows employees to query proprietary knowledge bases with high precision while maintaining data security via strict multi-role authentication protocols.

To support decision-making, the platform features a dynamic **Performance Analytics Dashboard** powered by Chart.js. This module visualizes KPIs and employee metrics, enabling leadership to gain actionable insights into talent performance trends. The dashboard is seamlessly integrated with the platform's role-based access control (RBAC), ensuring sensitive data is only accessible to authorized personnel.

The **AI Exam & 6-Day Study Sprint Orchestrator** automates the learning cycle. By analyzing competency gaps, the system auto-generates tailored exam content and structures accelerated 6-day learning sprints. This feature provides a scalable, personalized learning path for employees, directly mapping company training goals to individual career progression.

Interaction is revolutionized by the **Sphere Voice AI Assistant**. This module enables hands-free navigation and querying, incorporating advanced Speech-to-Text (STT) and Text-to-Speech (TTS) capabilities. Sphere utilizes sophisticated tool calling to interface with the backend, allowing users to initiate complex actions—like scheduling a training review or querying a performance metric—using natural language.

Finally, the backend leverages **FastAPI and Uvicorn ASGI** to provide high-performance, asynchronous endpoints. The integration of **SQLite WAL (Write-Ahead Logging)** mode ensures that read and write operations do not block each other, delivering robust concurrency management essential for enterprise multi-user environments.

4. Timeline Overview

Week	Activities Planned	Activities Completed
Week 1	Requirement analysis, Database schema design, Auth setup	Initialized SQLite WAL architecture and Multi-role Auth
Week 2	Document ingestion engine and RAG pipeline	Integrated BGE-Large embeddings & ChromaDB vector store
Week 3	Analytics dashboard & Performance metrics	Developed Chart.js dashboard integration
Week 4	Exam generation & Sprint orchestrator logic	Built AI-driven study sprint generator
Week 5	Sphere Voice AI Assistant development	Implemented Web Speech API/TTS integration
Week 6	System stress testing & Deployment	Finalized FastAPI/ASGI concurrency hardening

5. Key Milestones & Project Execution Details

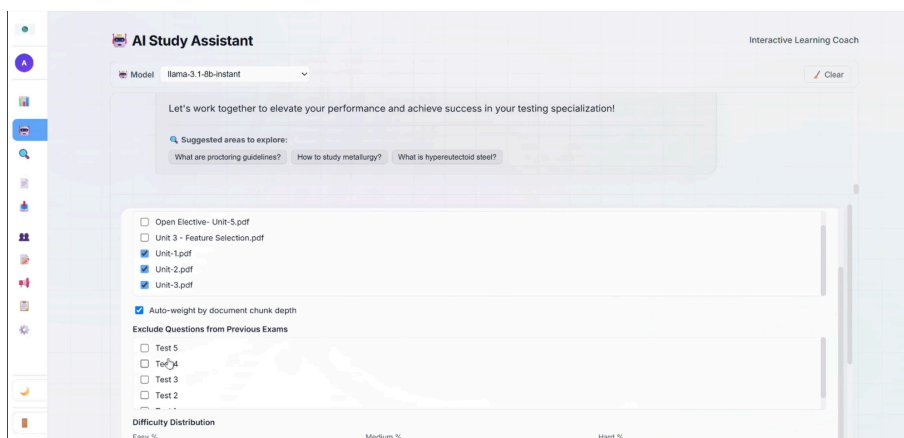
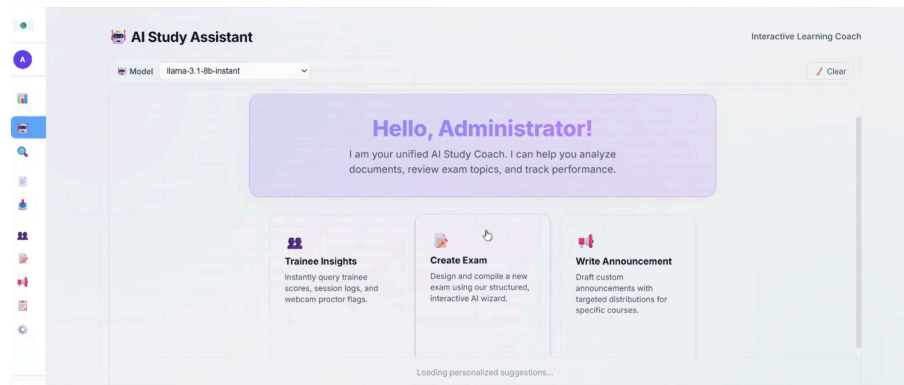
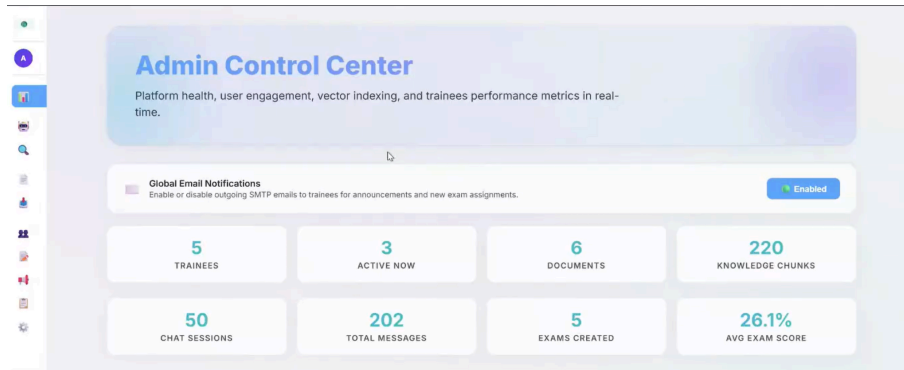
5a. Key Milestones

Milestone	Description	Status
Milestone 1	Document Ingestion, Vector Indexing & Retrieval	Completed
Milestone 2	RAG Chatbot, Talent Portal, AI Exams & Announcements	Completed
Milestone 3	Sphere Voice AI Assistant & AI Study Sprints	Completed
Milestone 4	Automated Notifications, DB-Aware Intelligence & Concurrency	Completed

5b. Project Execution Details

The project followed an agile methodology, prioritizing modular development to manage the complexity of integrating AI services with traditional enterprise backends. We utilized a **Flask MVC design** pattern to decouple the UI from the underlying business logic, which was later optimized for high-performance service delivery via **FastAPI** and **Uvicorn ASGI**. Database integrity was a critical focus; we implemented **SQLite WAL mode** to handle concurrent read/write requests, effectively eliminating file locking issues during high-load periods. Each module, from the RAG pipeline to the Voice AI Assistant, underwent rigorous unit testing to ensure the hybrid architecture remained stable and performant.

6. Snapshots / Screenshots



AI Study Assistant Interactive Learning Coach

Model: llama-3.1-8b-instant Clear

Trainee Insights
Instantly query trainee scores, session logs, and webcam proctor flags.

Create Exam
Design and compile a new exam using our structured, interactive AI wizard.

Write Announcement
Draft custom announcements with targeted distributions for specific courses.

Interactive Announcement Wizard 25%

Step 1 of 4: Title & Category

Announcement Title

Category: General Notice Priority: Standard

[Next: Edit Content](#)

My Assessments
View your assigned exams, complete tests, and review AI-graded comments.

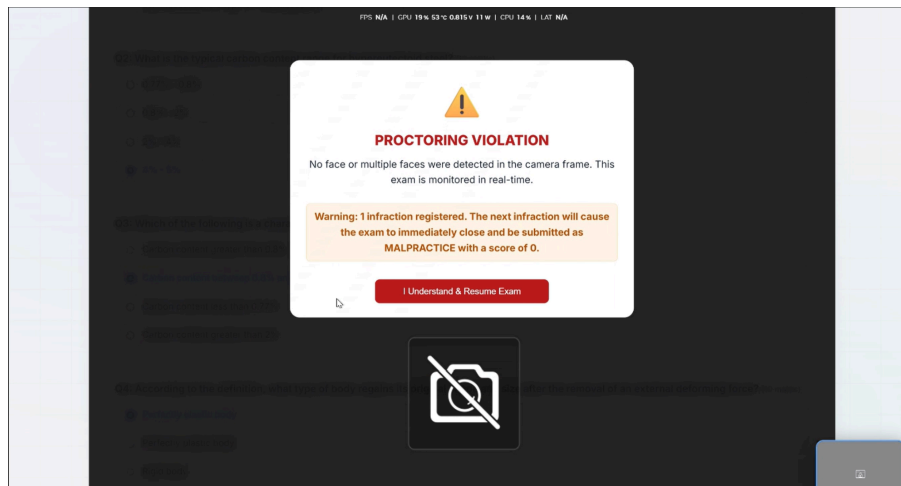
Assigned Exams

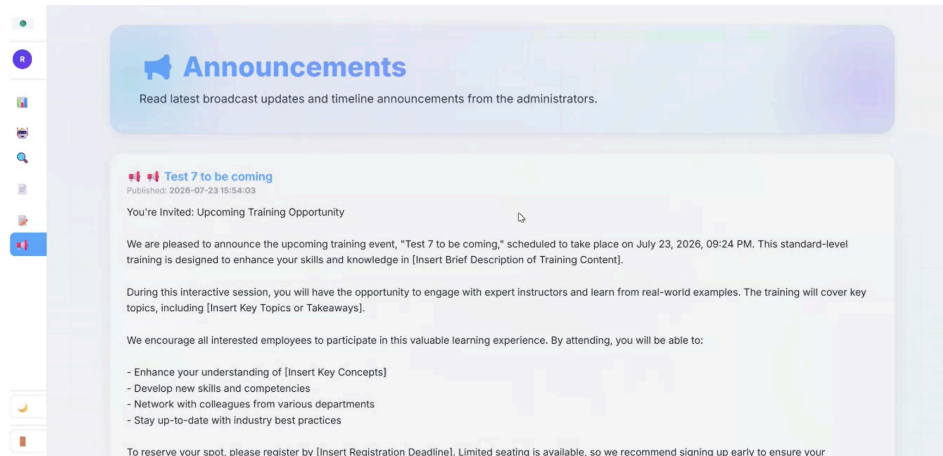
Test 6
Due Date: 2026-07-24T15:51 - Total Marks: 20
[Duration: 30 minutes] All the best [Take Exam](#)

Test 5
Due Date: 2026-07-23T17:45 - Total Marks: 40
[Duration: 30 minutes] All the best [Take Exam](#)

Completed Exams

Test 4
Submitted: 2026-07-22 13:29:44 - Score: 0.0 / 30 [Review Grade](#)





Announcements

Read latest broadcast updates and timeline announcements from the administrators.

Test 7 to be coming
Published: 2026-07-23 15:54:03

You're invited: Upcoming Training Opportunity

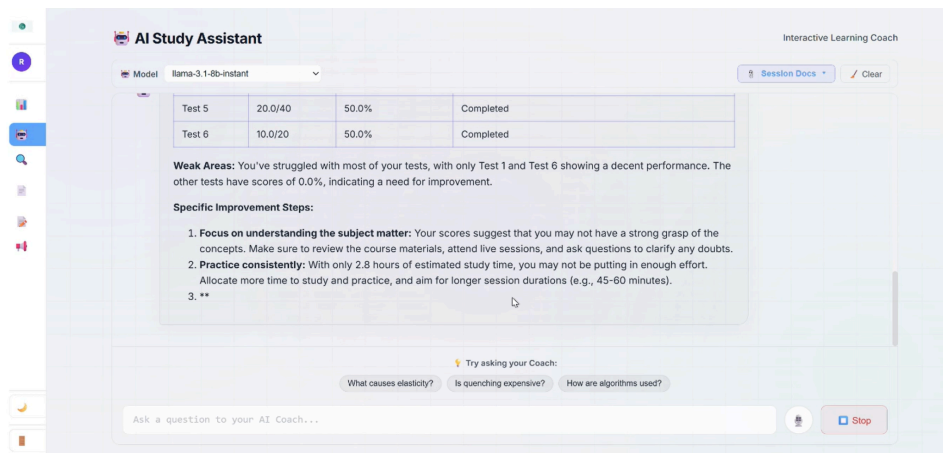
We are pleased to announce the upcoming training event, "Test 7 to be coming," scheduled to take place on July 23, 2026, 09:24 PM. This standard-level training is designed to enhance your skills and knowledge in [Insert Brief Description of Training Content].

During this interactive session, you will have the opportunity to engage with expert instructors and learn from real-world examples. The training will cover key topics, including [Insert Key Topics or Takeaways].

We encourage all interested employees to participate in this valuable learning experience. By attending, you will be able to:

- Enhance your understanding of [Insert Key Concepts]
- Develop new skills and competencies
- Network with colleagues from various departments
- Stay up-to-date with industry best practices

To reserve your spot, please register by [Insert Registration Deadline]. Limited seating is available, so we recommend signing up early to ensure your



AI Study Assistant Interactive Learning Coach

Model: llama-3.1-8b-instant

Test	Score	Percentage	Status
Test 5	20.0/40	50.0%	Completed
Test 6	10.0/20	50.0%	Completed

Weak Areas: You've struggled with most of your tests, with only Test 1 and Test 6 showing a decent performance. The other tests have scores of 0.0%, indicating a need for improvement.

Specific Improvement Steps:

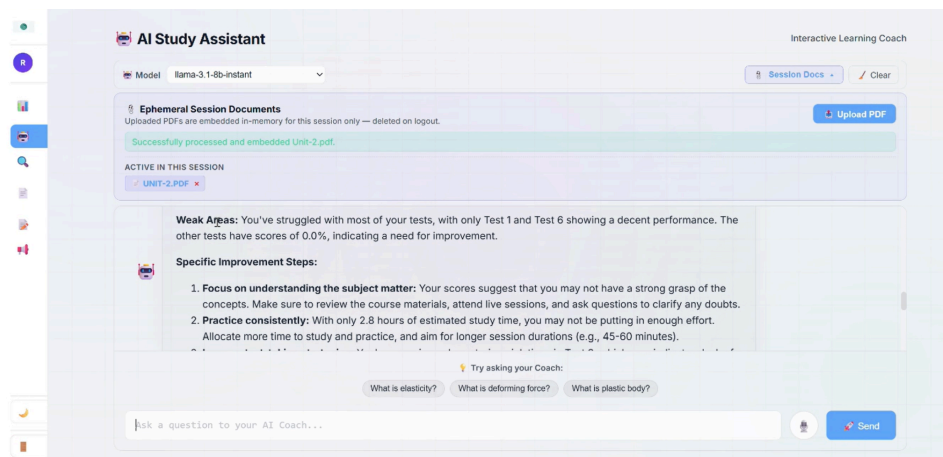
1. **Focus on understanding the subject matter:** Your scores suggest that you may not have a strong grasp of the concepts. Make sure to review the course materials, attend live sessions, and ask questions to clarify any doubts.
2. **Practice consistently:** With only 2.8 hours of estimated study time, you may not be putting in enough effort. Allocate more time to study and practice, and aim for longer session durations (e.g., 45-60 minutes).
3. **

Try asking your Coach:

What causes elasticity? Is quenching expensive? How are algorithms used?

Ask a question to your AI Coach...

Stop



AI Study Assistant Interactive Learning Coach

Model: llama-3.1-8b-instant

Ephemeral Session Documents
Uploaded PDFs are embedded in-memory for this session only — deleted on logout.

Successfully processed and embedded Unit-2.pdf.

ACTIVE IN THIS SESSION

UNIT-2.PDF

Weak Areas: You've struggled with most of your tests, with only Test 1 and Test 6 showing a decent performance. The other tests have scores of 0.0%, indicating a need for improvement.

Specific Improvement Steps:

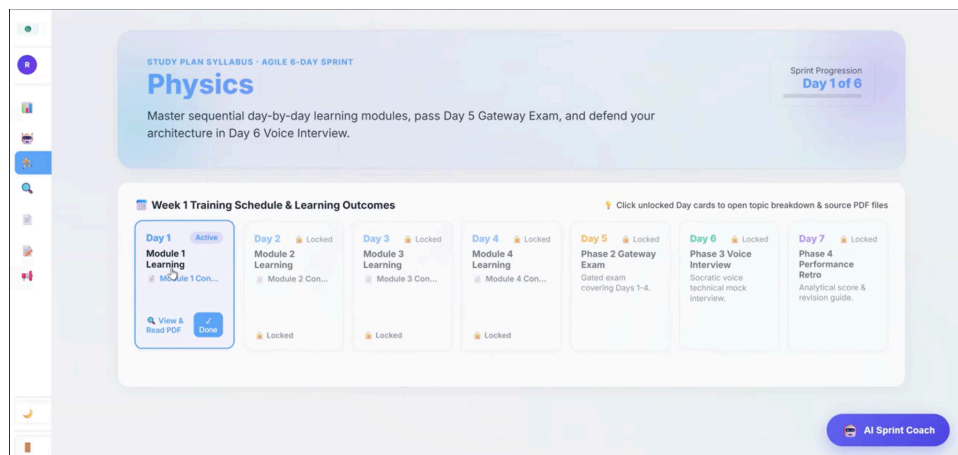
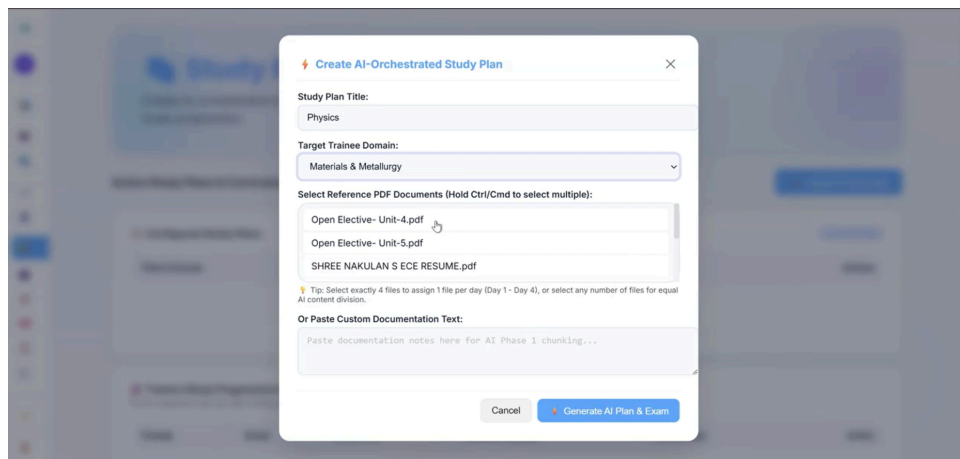
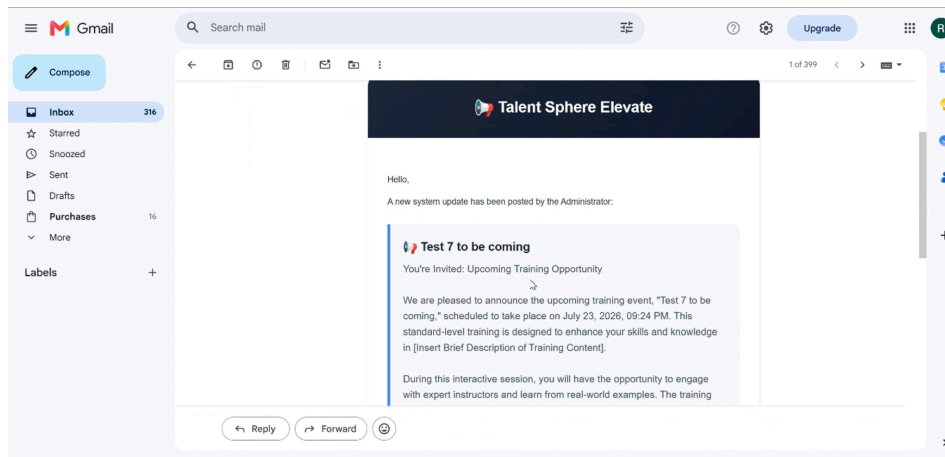
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2. **Practice consistently:** With only 2.8 hours of estimated study time, you may not be putting in enough effort. Allocate more time to study and practice, and aim for longer session durations (e.g., 45-60 minutes).

Try asking your Coach:

What is elasticity? What is deforming force? What is plastic body?

Ask a question to your AI Coach...

Send



7. Challenges Faced

- **SQLite Concurrency:** Initial development faced file-lock errors under load, which we successfully resolved by implementing SQLite's Write-Ahead Logging (WAL) mode, allowing simultaneous read/write operations.
- **ChromaDB Blocking:** High-frequency vector queries caused thread blocking in the main loop; this was solved by offloading inference tasks to a dedicated *ThreadPoolExecutor*.
- **Document Extraction:** We encountered significant noise when parsing complex scanned PDFs; we integrated a robust OCR fallback pipeline to maintain high-quality data ingestion.
- **Voice Timing Latency:** Synchronizing audio playback with AI processing events caused UI jitters. This was resolved using event-based synchronization hooks between the frontend and the Voice AI Assistant's backend.

8. Learnings & Skills Acquired

- **RAG/AI Pipeline:** Mastery of BGE-Large models, vector database architecture, and RAG optimization using ChromaDB.
- **Backend Engineering:** Advanced proficiency in FastAPI, Uvicorn ASGI, and asynchronous Python programming.
- **Database Optimization:** In-depth experience in SQLite WAL mode optimization and transactional data management.
- **Voice AI:** Practical implementation of Web Speech API and integrating LLMs for real-time natural language voice interactions.

9. Conclusion

The Talent Management Platform for Employee Performance and Career Growth represents a significant leap forward in enterprise learning and development. By successfully transforming static company documentation into actionable intelligence, the platform bridges the gap between raw data and employee career progression. The integration of RAG-based search, AI-orchestrated learning sprints, and real-time voice assistance creates a proactive ecosystem that empowers employees to own their growth while providing leadership with unprecedented visibility into talent metrics. This project demonstrates the tangible impact of applying enterprise AI to solve complex organizational challenges.